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Pay for performance? Government regulation and the structure of compensation contracts ☆

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Abstract

In 1992–1993, the SEC required enhanced disclosure on executive compensation and Congress enacted tax legislation limiting the deductibility of non-performance related compensation over one million dollars, i.e. Internal Revenue Code Section 162(m). We examine the effects of these regulatory changes and report small and large sample evidence that many million-dollar firms have reduced salaries in response to 162(m) and that salary growth rates have declined post-1993 for the firms most likely to be affected

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by the regulations. We further document that bonus and total compensation payouts are increasingly sensitive to stock returns after 1993, especially for firms with million-dollar pay packages. We also document that, once we control for other factors affecting CEO incentives, the sensitivity of the CEO's wealth to changes in shareholder wealth has increased from 1993 to 1996 for firms with CEOs near or above the million dollar compensation level. Overall, our results suggest that some firms have reduced salaries in response to 162(m). More importantly, the pay for performance sensitivity, measured using total annual compensation and firm-related CEO wealth, has increased for firms likely to be affected by 162(m).

JEL classificaion: G32; G34; G38; J33

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1. Introduction

CEO compensation has received recent public and academic scrutiny with most of the controversy focused on rising CEO compensation levels and on the absence of a strong relation between executive compensation and firm performance. As recently as February 25, 1999, Board of Governors' Chairman Alan Greenspan criticized the current executive compensation levels in Capitol Hill testimony on the state of monetary policy. Specifically, Greenspan said, "I find a lot of what is being paid to individual CEOs not directed to the value that they are producing for their shareholders, who are paying the bill."

Greenspan's concern is not new. Jensen and Murphy (1990), for example, analyze and discuss the low pay for performance sensitivity in the 1970s and 1980s. Recent evidence, suggests, however, that the pay for performance sensitivity may have increased since Jensen and Murphy studied the issue. Hall and Liebman (1998) document that the pay for performance sensitivity has increased since the early 1980s, with a striking acceleration in the last three years of their sample (1992–1994). They conclude that in recent years CEOs have not been paid like bureaucrats. One plausible explanation for this change is that the increasing pressure from outside investors on boards of directors has led to increased sensitivity of executive pay to firm performance. Johnson and Shackell (1997), Smith (1996), Strickland et al. (1996), and Wahal (1996), amongst others, document the rise in shareholder activism in the early 1990s.

The increased public attention on the pay for performance relation resulted in regulatory intervention by the SEC in 1993 requiring enhanced disclosure on executive compensation and the enactment of tax legislation limiting the deductibility of nonperformance related compensation over one million dollars

[Section 162(m) of the Internal Revenue Code, henceforth Section 162(m) or 162(m)]. The purpose of the new SEC disclosure requirements, as stated in the SEC's first release on July 2, 1992, was to make disclosure of compensation paid or awarded to executive officers clearer and more concise, and of greater utility to shareholders. Murphy (1995) explains that the SEC's 1992 proxy reform initiative was a response to the public outcry on executive compensation and a proposed Senate bill on shareholder rights and CEO compensation.

In addition, Congress never intended for Section 162(m) to be a revenue-raising provision, but instead Congress hoped to change corporate behavior.[1] When adopting 162(m), the House Ways and Means Committee stated the congressional intent in the following way:

> Recently, the amount of compensation received by corporate executives has been the subject of scrutiny and criticism. The committee believes that excessive compensation will be reduced if the deduction for compensation (other than performance-based compensation) paid to the top executives of publicly held corporations is limited to $1 million per year.[2]

Changing behavior could mean that the changes have led to real economic effects or that the compensation contracts are cosmetically different but that the economic effects are unchanged. Miller and Scholes (1982) examine the influence of personal taxes on compensation contracts and argue that some compensation plans are tax-advantageous schemes because they defer payments for employees in tax brackets that are higher than the corporation's tax bracket, therefore implying that personal taxes at the very least have a cosmetic effect on the design of compensation contracts. Hite and Long (1982) and Goolsbe (2000) also highlight how changes in tax laws can affect compensation policy.

In this paper, we examine whether the enhanced disclosure rules and 162(m) have actually changed compensation behavior. First, we identify firms from the *ExecuComp* database that are likely to be affected by the regulations and 162(m), henceforth generically called *million-dollar firms*. We define three variables to identify the million-dollar firms. The first variable is equal to one if the CEO's salary is more than one million dollars in the prior year, and is equal to the prior year's salary divided by one million dollars if the prior year's salary is less than one million dollars. Hall and Liebman (2000) use a similar definition to identify firms likely to be affected by 162(m). The second variable is an indicator variable equal to one if the firm's CEO earns a salary that is larger than $900,000 in the prior year, and zero otherwise. With this variable,

[1] Megan M. Reilly, 1994. Former Treasury Official Discusses Executive Compensation Cap, 62 Tax Notes 747, February 3.

[2] 1993 U.S.Code Congressional and Administrative News 877.

we intend to capture the firms that are already subject to the 162(m) limitation and the firms that are getting close enough to this benchmark to be concerned about loss of deductibility. Third, we define an indicator variable equal to one for firms with CEOs earning annual cash compensation (including salary and bonus) of more than one million dollars at least once over the 1992–1997 period. This broad definition includes about half of the firms in our sample suggesting that 162(m) is relevant for many firms. Overall, our results are qualitatively similar using the three definitions.

To examine whether and how these regulations influenced CEO compensation, we test five propositions which we formulate in our later sections. Our main findings are as follows: (1) Real compensation levels have increased dramatically in the period following the enactment of 162(m), in contrast to the stated intentions of Congress. Although rising stock option grants contribute greatly to these increases, all compensation components have increased in real terms and there is no evidence that the growth rates of the various components of compensation have declined after 1993. (2) Controlling for performance, salary growth has actually increased after 1993, but the growth rate is significantly smaller for firms near or above the million-dollar threshold. (3) For a subset of firms that reduce salaries to a level at or below one million dollars, we find that firms reduce salaries in response to 162(m). At the same time, these salary reductions do not typically lead to lower total compensation for the CEOs of these firms. (4) For all firms and especially those firms affected by 162(m), we find an increase in the sensitivity of bonus payments and total compensation to contemporaneous and lagged stock performance after 1993. This increased sensitivity of compensation to stock returns is not due to the mechanical increase in stock option values when the stock market is rising. (5) The performance sensitive components of compensation, especially stock option grants, have become much larger components of total compensation following 1993. Overall, these results suggest that compensation committees have taken 162(m) into account by modifying the structure of compensation contracts. Additional analysis of the changes in firm-specific CEO wealth and compensation suggests that these changes have also had a significant economic impact beyond annual compensation flows determined by the board of directors.

In the first part of our paper, we emphasize annual pay, defined as the annual flows of cash compensation and option and restricted stock grants. In the context of the regulations, it is important to analyze this annual flow because: (1) the compensation committee of the board of directors has a direct influence over this annual flow to the CEO; (2) shareholder activists, the press, regulators, and many previous academic studies have focused on the annual pay of CEOs and not on changes in the value of CEO equity holdings; and (3) the regulations were directly targeting this annual flow, especially cash compensation. We recognize, however, as documented by Jensen and Murphy

(1990) and Hall and Liebman (1998), that the explicit relation between CEO stock and option holdings and shareholder wealth provides most of the CEO incentives and we focus on this aspect in the second part of the paper. We follow Hall and Liebman by examining three measures of company-related CEO wealth changes and find that the sensitivity of CEO wealth to shareholder wealth is higher than the one reported by Jensen and Murphy for the 1970s and 1980s. For instance, using the Jensen-Murphy statistic or sharing rate, defined as the dollar change in CEO wealth for a $1,000 change in firm value, we document median CEO sharing rates in 1996 from $7 to about $25 per $1,000 shareholder gain or loss, depending on the subgroup. This is in line with the Hall and Liebman finding that pay to performance sensitivity in the 1990s is substantially higher than the sensitivity documented by Jensen and Murphy for earlier periods. More importantly, in a regression framework we find that there is an increase in the wealth to performance sensitivity from 1993 to 1996 for firms approaching the million-dollar benchmark, i.e. firms affected by 162(m).

The paper is organized as follows. In Section 2, we discuss the background for the changes in the SEC Compensation Disclosure Rules and 162(m). In Section 3, we discuss our data collection procedure. Section 4 presents our empirical results on the effects of the regulation on compensation structure, salary levels and growth rates, and compensation to performance sensitivity. Section 5 evaluates the effects of the regulations on the sharing rate and other measures of the sensitivity of CEO wealth on changes in shareholder wealth. We examine the robustness of our results in Section 6 and offer concluding comments in Section 7.

2. The SEC's new Compensation Disclosure Rules and I.R.C. Section 162(m)

As a result of public perception regarding excessive executive pay in the early 1990s, executive compensation became a hot political issue. As a result, the Securities and Exchange Commission adopted new rules regarding disclosure of executive compensation in 1992.[3] These rules required that corporations enhance the disclosure of executive compensation in proxy statements beginning with the 1993 proxy season. Under the new rules, firms are required to: (1) compare their financial performance to an industry benchmark with a table and performance graphs; (2) disclose in a tabular form the annual and long-term compensation for the CEO and the four most highly paid executives; (3) provide estimates of the present value of managerial stock options granted; and (4) provide a report by the compensation committee explicitly identifying quantitative or qualitative performance measures used to evaluate managers.

[3] Securities and Exchange Commission, Release Nos. 33-6962; 33-6966; 33-7009; and 34-32723.

Researchers have examined managerial reporting discretion relating to portions of these rules and their results suggest that managers and/or directors place great importance on compensation disclosure. For example, Murphy (1996) examines the limited discretion given managers in valuing stock options and finds that managers tend to select methodologies that reduce reported compensation. He concludes that managers bear non-pecuniary costs from high reported levels of compensation. Along the same vein, Byrd et al. (1998) and Lewellen et al. (1996) report that managers choose industry- and peer-company stock return benchmarks that are downward biased. Thus, they argue that managers pay attention to the compensation disclosure as they choose data that tend to overstate the firm's performance.

As part of the Omnibus Budget Reconciliation Act of 1993, Congress enacted Section 162(m) of the Internal Revenue Code (1995), a provision that disallows deductions for nonperformance related compensation over one million dollars for the CEO and the other executives whose compensation must be reported in the proxy statement. The applicable employee remuneration does not include remuneration payable on a commission basis, compensation that is performance-based, and compensation under a binding written contract in effect on February 13, 1993. For the purpose of a deduction, compensation is classified as performance-based only if (i) the performance goals are determined by a compensation committee comprised solely of two or more outside directors, (ii) the performance goals under which the remuneration is to be paid are disclosed to the shareholders and approved by a majority vote, and (iii) before any payment of such remuneration, the compensation committee certifies that the performance goals and other material terms were satisfied.

As stated in our introduction, the intention of Congress in enacting 162(m) was not to raise substantial revenue for the government, but to shape corporate behavior. For example, a July 1993 projection of the Joint Committee on Taxation estimated total revenues generated from 162(m) over 1994–1998 to be only $335 million. In comparison, the simultaneous reduction of business meal deductions from 80% to 50% was expected to raise $15.4 billion over the same period. We gauge the potential direct economic impact of non-deductibility by estimating the potential loss of tax shield for each firm arising from cash compensation in excess of one million dollars. Specifically, we compute the salary and bonus in excess of one million dollars for each of the named executive officers of the firm reported on *ExecuComp* and multiply this excess compensation by the firm's estimated marginal tax rate and divide it by net income. To estimate the marginal tax rate, we use the trichotomous variable as discussed by Graham (1996) and Graham and Lemmon (1998). For 1994–1997, the median potential loss of tax shield for firms with at least one executive officer (including the CEO) receiving cash compensation in excess of one million dollars is only 0.06% of net income. Thus, for most firms, nondeductibility of excess compensation would not significantly affect net income.

Existing research on 162(m) suggests that 162(m) has affected the way firms approach compensation issues. Livingstone (1997) documents that many compensation committees are careful to maintain deductibility of executive pay under 162(m) and Anderson and Bizjak (2000) argue that the new SEC disclosure requirements and 162(m) increase the pressure to have compensation committees consisting of independent directors. A Towers Perrin survey described in the *Investor's Business Daily* of March 10, 1995 reports that 87% of the firms surveyed intended to implement Section 162(m) changes for positive shareholder relations while only 43% of the firms mentioned that financial (tax) considerations were important.

3. Data collection

We retrieve compensation data from Standard and Poor's *ExecuComp* database. This database includes S&P 500, Midcap, and Smallcap index firms. Compensation data are classified into the seven compensation categories identified in the Summary Compensation Tables of the proxy statements including salary, bonus, other annual compensation (including perquisites and amounts for reimbursed for payment of taxes), restricted stock awards, options or stocks appreciation rights (SARs), and long-term incentive plan payouts (LTIP), and "all other compensation." Total compensation consists of all seven components reported in the proxy statement. The Disclosure Rules require firms to provide dollar values in the Summary Compensation Table for each of the components, except options and SARs, and *ExecuComp* reports the dollar values provided in the proxy statements. For options and SARs, firms are only required to disclose the number granted to their top executives in the Summary Compensation Table, however additional information relating to the stock option grants is available elsewhere in the proxy statement and *ExecuComp* separately computes the present value of the options granted to executives using a modified Black-Scholes method. We retrieve accounting data from *Compustat* and stock returns from the *CRSP* files. For a smaller subset of our firms, we also collect information on the performance measures explicitly used by compensation committees to evaluate the CEO from the compensation committee reports in the proxy statements. For several of our tests discussed in the sections below, we compare compensation before 162(m) to compensation after 162(m). For the pre-162(m) period, which we identify with the post-1993 dummy variable in our tests, we only have 1991, 1992, and 1993 compensation data that are truly comparable to the post-162(m) data. This data limitation reduces the power of our tests.

4. The effect of the regulations on annual CEO compensation levels, structure, and relation with performance: propositions and tests

As reported in previous sections, the main objective of Congress in adopting Section 162(m) was to reduce excessive compensation. Thus, we formulate

Proposition 1. CEO compensation levels decrease following the adoption of 162(m) and the SEC disclosure changes.

Henceforth, we refer to 162(m) and the SEC disclosure as "the regulations" and use 1993 as a key year because these regulations were adopted in late 1992 and early 1993 and became effective after 1993. To examine our Proposition 1, we analyze the mean, the median (both in CPI-deflated 1992 constant dollars), and the number of observations for the six main compensation components for 1992–1997 in Table 1.

Over the sample period, all compensation components have increased dramatically even in real terms. The median salary, the compensation component assumed to be least sensitive to performance, has increased by 17%. Bonuses, LTIP payouts, and grants of restricted stock have all nearly doubled from 1992 to 1997. The present value of option grants has trebled in real terms over that period.[4] Overall, the data in Table 1 suggest a continuation, or even acceleration, of the rising compensation trend and the increased use of stock options already documented by Hall and Liebman (1998) and Murphy (1999). Clearly, the regulatory changes have not curbed overall CEO compensation levels.

4.1. Firms likely to be affected by the regulations

While the SEC compensation disclosure requirements affect all firms, 162(m) does not directly affect about one half of the firms in our sample, i.e. the firms with cash compensation lower than one million dollars. In the following subsections, we examine how the regulations have affected a subset of firms that are likely to be more sensitive to these regulations. We first focus on CEO salaries because salaries are by definition not explicitly tied to performance and are subject to the 162(m) limitation. The threshold for deductibility is a nominal $1,000,000, hence salaries are not CPI-adjusted for these tests.

[4] The 1992 options sample is not comparable to the 1993–1997 sample because *ExecuComp* reports option values for only 959 CEOs in 1992. Firms with option information for 1992 are generally larger firms with larger option grants.

Table 1
Descriptive statistics on CEO compensation and ownership for *ExecuComp* firms over 1992–1997

We report mean and median compensation data (in CPI-deflated 1992 constant dollars) for the CEOs in the *ExecuComp* data set. The figures contain the six compensation components from the proxy statements; salary, bonus, other annual compensation, long-term incentive plan payouts, managerial stock options as valued by *ExecuComp*, and restricted stock, as well as the sum of all six components, i.e. total compensation. Total compensation is only presented when option values are available for the observation. We also include percentage ownership. Stock ownership is the number of shares divided by the number of shares outstanding. Option ownership is the number of exercisable and unexercisable options divided by the number of shares outstanding.

		1992	1993	1994	1995	1996	1997
Salary (000s)	Mean	480	480	494	505	514	549
	Median	426	420	437	450	463	499
	N	1,271	1,495	1,571	1,565	1,551	1,213
Bonus (000s)	Mean	332	377	437	479	571	623
	Median	200	199	249	243	277	341
	N	1,271	1,495	1,571	1,565	1,551	1,213
Other annual compensation (000s)	Mean	37	28	29	32	39	38
	Median	0	0	0	0	0	0
	N	1,078	1,495	1,571	1,565	1,551	1,213
Long-term incentive plans (000s)	Mean	91	78	69	113	161	184
	Median	0	0	0	0	0	0
	N	1,271	1,495	1,571	1,565	1,551	1,213
Options (000s)	Mean	615	520	671	743	1,238	1,904
	Median	184	143	176	196	338	485
	N	959	1,417	1,446	1,480	1,543	1,209
Restricted stock (000s)	Mean	114	108	117	142	185	240
	Median	0	0	0	0	0	0
	N	1,271	1,495	1,571	1,565	1,551	1,213
Total compensation (000s)	Mean	1,770	1,672	1,895	2,089	2,797	3,747
	Median	1,078	1,049	1,134	1,192	1,445	1,806
	N	949	1,417	1,446	1,480	1,543	1,209
Stock ownership	Mean	3.9%	3.9%	3.6%	3.5%	3.3%	3.0%
	Median	0.4%	0.6%	0.5%	0.5%	0.5%	0.4%
	N	994	1,361	1,415	1,419	1,453	1,152
Option ownership	Mean	1.0%	0.9%	1.0%	1.0%	1.1%	1.1%
	Median	0.4%	0.4%	0.5%	0.6%	0.6%	0.7%
	N	957	1,368	1,422	1,422	1,461	1,157

4.1.1. Increases versus decreases in salary: univariate statistics

Table 1 shows that salaries have not decreased as a whole but it is conceivable that the regulations have been effective in putting downward salary pressure on firms with million-dollar CEO salaries. Thus, we formulate

Proposition 2A. Salaries above or nearing the million-dollar range are less likely to increase than salaries farther below the million-dollar mark.

To provide a first examination of this proposition, we categorize firms in six subsets based on salary in the prior year and examine what proportion of firms in each category increased, decreased, or did not change the CEO's salary and report these statistics in Table 2.[5] In the categories below \$900,000, between 75% and 84% of the salaries increase post-1993. In contrast, in the categories above \$1,000,000, only between 37% and 72% of salaries are increased in the post-1993 years. There is a general tendency for high salaries to be less likely to increase than lower salaries, especially for salaries between one million dollars and 1.1 million dollars. These univariate results suggest that in the subsets where firms are more likely to be affected by 162(m), the proportion of salary increases is significantly lower after 1993 than in the subsets with lower salaries. Still, the number of CEOs earning more than one million dollars increased from 25 (3.5% of the firms) in 1992 to 81 (7.4% of the firms) in 1997. Thus, while lower salaries are more likely to increase than higher salaries, our results confirm our prior findings that the regulations have not succeeded in reducing overall salary levels. This analysis is suggestive only and does not explain whether we observe this tendency because of reversion to the mean or whether this tendency is related to the size and performance characteristics of the high salary firms. We investigate these aspects in the next few subsections.

4.1.2. Why do firms reduce million-dollar salaries?

Results in the preceding section show that a significant fraction of high salaries are reduced in any one year. To test whether these reductions result from changes in the regulations, we formulate

Proposition 2B. Firms reduce salaries above one million dollars because of 162(m).

To directly test this proposition, we identify 164 firms with salaries over one million dollars at any time over 1992–1996. Of those 164 firms, 44 reduce the CEO salary to one million dollars or below in the next year. We identify the stated reasons for this reduction by examining the Compensation Committee Report in the proxy statement for each of the 44 firms. For 19 of the 44 firms, the salary is reduced following CEO turnover. The 25 remaining firms are put in five categories provided in Table 3. Whereas only one firm explicitly highlights the importance of poor performance, 23 of the 25 firms that reduce the salary to one million dollars or less highlight the importance of 162(m) in

[5] We use a variety of cutoff points to examine salary changes across salary levels. The results are similar in nature if we use different cutoff points.

Table 2
Salary levels and changes in salary for *ExecuComp* firms over 1992–1997

This table documents the number of *ExecuComp* within certain compensation ranges, and for each salary range, the percentage of salaries that have decreased, not changed, or increased, compared to the base at the beginning of that year.

Salary levels		1992	1993	1994	1995	1996	1997
Less than or equal to 499,999	*N*	405	738	803	783	730	503
	Decrease (%)	3.46	5.28	4.48	4.21	5.89	2.19
	No change (%)	15.80	15.18	13.33	15.45	14.93	14.31
	Increase (%)	80.74	79.54	82.19	80.33	79.18	83.50
500,000–699,999	*N*	196	258	281	306	344	295
	Decrease (%)	5.10	5.81	3.56	3.59	4.65	3.39
	No change (%)	17.86	15.50	18.15	14.38	18.02	14.24
	Increase (%)	77.04	78.68	78.29	82.03	77.33	82.37
700,000–899,999	*N*	80	146	155	160	157	157
	Decrease (%)	1.25	7.53	4.52	2.50	3.82	1.91
	No change (%)	26.25	15.07	18.71	18.75	21.02	19.11
	Increase (%)	72.50	77.40	76.77	78.75	75.16	78.98
900,000–999,999	*N*	18	24	37	45	51	60
	Decrease (%)	0.00	12.50	5.41	4.44	7.84	6.67
	No change (%)	5.56	12.50	13.51	28.89	23.53	26.67
	Increase (%)	94.44	75.00	81.08	66.67	68.63	66.67
1,000,000–1,099,999	*N*	9	19	23	29	39	41
	Decrease (%)	22.22	26.32	17.39	10.34	2.56	17.07
	No change (%)	11.11	15.79	26.09	51.72	46.15	46.34
	Increase (%)	66.67	57.89	56.52	37.93	51.28	36.59
Greater than 1,099,999	*N*	16	28	35	36	39	40
	Decrease (%)	6.25	10.71	20.00	16.67	17.95	10.00
	No change (%)	25.00	25.00	22.86	27.78	20.51	17.50
	Increase (%)	68.75	64.29	57.14	55.56	61.54	72.50
Total for all firms	*N*	724	1,213	1,334	1,359	1,360	1,096
	Decrease (%)	3.87	6.27	4.95	4.34	5.66	3.56
	No change (%)	17.40	15.42	15.44	17.14	17.79	16.97
	Increase (%)	78.73	78.32	79.61	78.51	76.54	79.47

determining CEO salaries. For example, the compensation committee's report in Iowa Beef Processors' 1995 proxy statement states:

> The Chairman and Chief Executive Officer's salary and performance-based bonus for 1994 were established by the Compensation Committee in December of 1993. Mr. Peterson's base salary was decreased to $1,000,000 from $1,240,000.... . These actions were based on the changes to Section

Table 3
Categorization of the various (non-CEO turnover related) reasons provided in proxy statements for CEO salary reductions from over one million dollars over 1992–1996 to at or below one million dollars thereafter.

Category	Number of firms	Percentage
1. Salary reduction because of poor performance	1	4.0
2. Salary reduction in connection with a merger	1	4.0
3. Reasons related to 162(m)-		
Deferral of compensation over $1 million for 162(m)	1	4.0
Complying with 162(m)	16	64.0
May defer compensation or modify contracts to comply with 162(m) but reserve the right to make payments that would not qualify for deduction under 162(m)	6	24.0

> 162(m) of the Internal Revenue Code which requires that any compensation over $1,000,000 be performance-based (or meet other exceptions provided by the Section) to be deductible by the Company. The salary and performance-based bonus were determined pursuant to the changes to Section 162(m) and in order to retain Mr. Peterson as Chairman and Chief Executive Officer. The bonus method was designed to incentivize Mr. Peterson with a performance-based bonus that was competitive with the industry and also allows the Company to take a deduction for federal income tax purposes.

Our results so far suggest that firms with CEO earning salaries close to or over one million dollars are more likely to cut salaries than firms with lower CEO salaries and that when million-dollar salaries are cut, most firms attribute the reduction to changes in 162(m). While these results suggest that the regulations have an effect on the salary levels of some million-dollar firms, a majority of million-dollar firms still increase rather than reduce salaries. Further, of the 25 firms reducing salaries below one million dollars, only eight firms actually reduce total annual compensation to the CEO in that year. In fact, the median change in total CEO compensation for the 25 firms that reduce salaries is an increase of 28% the year that the salary is reduced. This finding reinforces our prior result that 162(m) did not reduce (the increases in) total compensation, even for firms that cut salaries as a result of 162(m). We examine this issue in more detail in Section 4.2.[6]

[6] In our later tests, we also analyze this sample of firms separately to examine how their pay for performance relation changes over time. Unfortunately, the small sample size does not allow us to draw conclusions on the pay for performance relation of this sample of firms that are significantly different from our conclusions for the overall sample. We also examine whether this subsample drives our large sample results by reestimating Tables 5, 6 and 8 without these observations, and the omission of these observations does not change the substance of the reported results.

4.2. Salary growth rates: multivariate analysis

Our results in Section 4.1 suggest that firms nearing the million-dollar range are less likely to increase the CEO's salary than firms with lower salaries. Specifically, the results in Section 4.1.2 indicate that compensation committees report that salary reductions for million-dollar firms (in this case CEOs effectively earning salaries of more than one million dollars) are in most cases due to 162(m). Hence, we formulate and test

Proposition 2C. Salaries nearing the million-dollar range increase less than salaries further below the million-dollar range.

To have a better feel for the data, we first examine, but do not tabulate, CEO salary growth rates prior to 1994 and after 1993. Prior to 1994, the median salary growth rate for CEOs earning less than $900,000 is 5.8% compared to 4.6% for CEOs earning more than $900,000. After 1993, the growth rate for CEOs with lower salaries accelerates to 6.0%, whereas the median growth rate for CEOs with salaries above $900,000 declines to 2.7%. The acceleration in growth rates for lower-paid CEOs and the decline in growth rates for higher-paid CEOs are both statistically different from zero. The salary growth rates for the two subgroups are also significantly different from one another after 1993, indicating that salary growth rates are lower for CEOs with salaries close to or beyond the million-dollar benchmark. The sections below examine this issue more formally by including control variables known to impact CEO compensation levels.

4.2.1. Control variables likely to affect compensation levels

Firm performance is likely to affect salary levels through time. Both in compensation practice and in the academic literature, there have been differences in opinion regarding the best measure of performance. In the finance literature, Murphy (1985), Lambert and Larcker (1987), Ely (1991), Paul (1992), and Sloan (1993), among others, have used stock returns, accounting returns, and sales, as performance variables. We take a slightly different approach to identify relevant performance variables. This approach, now possible with the improved compensation disclosure, offers additional insights into the pay and performance relation. We have selected a random sample of 100 S&P 500 and 100 Midcap 400 firms and document the performance measures identified by their compensation committees as the criterion used to determine CEO compensation including bonuses. The results of this analysis are summarized in Table 4. At least 97% of the firms use some form of financial performance measure in determining the CEO bonus. To better understand which type of performance measures are prevalent, we consolidate the various measures into categories. For example, 60% of the

Table 4
Distribution of performance variables used by compensation committees to determine the CEO bonus

This table presents the main financial and non-financial performance measures used by compensation committees for a sample of 100 randomly selected S&P 500 and 100 randomly selected Midcap 400 firms. The measures are retrieved from the 1995 proxy statements.

Panel A: Financial performance measures	All	S&P	Midcap	
% of firms using financial measures	97	99	95	
% of firms using measures based on:				Examples
Net income	60	51	69	Net income, net profit, net income minus non-recurring events
EPS	38	41	34	Change in EPS, EPS
Sales	32	30	34	Sales, revenue
ROE	31	34	28	Net ROE, operating ROE, EVA, increase in book value per share
Shareholder returns	20	19	20	Shareholder returns, returns versus peers
ROA	15	14	15	Net ROA, Operating ROA
Cash flows	15	15	14	Cash flows, free cash flows, improvement in operating cash flows
Profit margin	13	14	11	Return on sales, profit margin, expense ratio
Dividends	2	1	2	Increase in dividend payout
Panel B: Non-financial performance measures				
% of firms using non-financial measures	69	64	73	
% of firms using measures based on:				Examples
Individual business objectives	31	31	30	Safety, market share, acquisitions, spin-offs, restructuring, selling assets, growth
Strategic targets	22	17	26	Strategic positioning, corporate objectives
Leadership	16	15	17	Provide leadership
Customer satisfaction	18	15	21	Customer satisfaction, service quality
Human resources	12	11	13	Human resources development

firms mention net income, net profit, or profit minus nonrecurring events, which we lump together in the net income category. The next most frequently used financial measures, in order, are earnings per share (EPS), sales, return on equity (ROE), shareholder returns, cash flows, return on assets (ROA), profit margin, and dividends. A variety of measures are used by only one firm in the subsample, such as revenue per kilowatt-hour. We also document that 69% of the firms use nonfinancial performance measures such as meeting individual business objectives or incorporating the leadership role of the CEO.

Ideally, we would like to replicate the exact compensation formulae for each firm and use this formula in our analysis of the pay for performance sensitivity. The reports rarely explain, however, the exact methodology used by firms to calculate these measures or reveal the weight each financial measure receives and the quantitative threshold values and benchmarks. Based on previous research and the results reported in Table 4, we choose three performance measures that are likely to capture the information in the most commonly used performance measures: EPS, sales, and stock returns. EPS is a net income measure that is not as correlated with firm size as basic net income. To facilitate comparison with prior studies, we scale EPS by the previous period's fiscal year-end stock price. We use sales because it is still used frequently in bonus programs, and because prior compensation studies document its strong relation with salary levels. Finally, we use total return to shareholders because it is another commonly used performance measure that is less correlated with sales and EPS, and it is the performance measure that is used predominantly by practitioners, academic researchers, and investors.

We also need to discuss the timing of the performance variables. Salaries are negotiated before the year's results are known, so the prior year's performance is likely to affect the current year's salary. The situation is different for the bonus, where this year's performance typically affects the current year's bonus. The timing of when a company reports bonuses can vary depending on which fiscal year the bonus is actually received by the CEO. If an annual bonus is paid subsequent to the close of the fiscal year, the bonus may be reported in the following year's compensation, resulting in a lag similar to the one with salaries. To ensure that we include the relevant performance variables, we include both lagged and contemporaneous performance in all our models.

As additional control variables, we include measures of CEO tenure and firm size. Murphy (1986), among others, demonstrates that tenure affects the structure of compensation contracts. Smith and Watts (1992) and Coughlan and Schmidt (1985), among others, highlight the importance of firm size in determining executive compensation. We use the logarithm of total assets of the firm to control for firm size. Other firm specific and industry specific variables may also affect compensation, but we control for their likely influence on compensation levels by controlling for firm effects in our analysis of the sensitivity of bonus payments and total compensation to performance.

4.2.2. Multivariate model

To examine changes in salary, we estimate the following regression model:

$$\begin{aligned}\Delta \mathrm{Ln}(\mathit{Salary}_{it}) = {} & \alpha_0 + \alpha_1 \mathrm{Ln}(\mathit{CEOtenure}_{it}) + \alpha_2 \mathrm{Ln}(\mathit{Assets}_{it}) \\ & + \sum_{j=3}^{8} \alpha_j \mathit{Performance\ Measures}_i \\ & + \alpha_9 \mathit{Million\text{-}dollar\ variable} \\ & + \alpha_{10} \mathit{Million\text{-}dollar\ variable} * \mathit{Post93dummy} \\ & + \mathit{Yeardummies} + \varepsilon_{it}. \end{aligned} \tag{1}$$

The model's independent variables are:

$\mathrm{Ln}(\mathit{CEO\ tenure}_{it)})$	= the natural logarithm of the number of years the current CEO of firm i has held the position of CEO in year t
$\mathrm{Ln}(\mathit{Assets}_{it)})$	= the natural logarithm of firm i's contemporaneous assets in year t
Million-dollar variable	= (1) a semi-continuous variable equal to one if firm i's CEO's salary was equal to or greater than one million dollars in year $t-1$ and equal to the CEO's salary divided by one million dollars if the CEO's salary was less than one million dollars (see Hall and Liebman, 2000), or (2) a dummy variable equal to one if the CEO earned a salary of more than \$900,000 in the preceding year, or (3) a dummy variable equal to one for firms with CEOs earning annual cash compensation of more than one million dollars at least once over the 1992–1997 period
Post93dummy	= a dummy variable equal to one for compensation in 1994, 1995, 1996 or 1997 and equal to zero for 1992 and 1993
Performance measures	
EPS_{it}, EPS_{it-1}	= earnings per share, excluding extraordinary items and discontinued operations, for years t and $t-1$ for firm i, scaled by the previous period's fiscal year-end stock price

ΔLn(*Sales*$_{it}$), ΔLn(*Sales*$_{it-1}$)	= the change in the logarithm of sales for years t and $t-1$ for firm i
Holding period return$_{it}$ *Holding period return*$_{it-1}$	= holding period returns for years t and $t-1$ for firm i

4.2.3. Results

Table 5 reports the results of our analysis on changes in salary. We present quantile regression results with bootstrapped standard errors (*bsqreg* in STATA). In model (i), we use the semi-continuous million-dollar variable definition; in model (ii) we use a dummy variable equal to one for CEOs with salaries higher than \$900,000 in the preceding year as the million dollar variable; and in model (iii) we use a dummy variable equal to one if the firm had a CEO that earned more than one million dollars in salary and bonus at least once over the sample period. An important distinction between the definition of million-dollar firms in the first two models and the third model is that the value of the third variable does not change for a firm over the sample period, whereas the value of the first two variables can change over time.

Overall, longer tenures are associated with smaller increases in salary and in the first model, size is positively related to changes in salary. In addition, contemporaneous earnings and returns are unrelated to changes in salary in all three models, whereas contemporaneous and lagged sales as well as lagged earnings (in the last two models) and lagged returns have a positive effect on changes in salary. The importance of lagged performance measures over contemporaneous performance measures is expected given the way salaries are set, i.e., before the contemporaneous year's performance is known. Changes in salary are negatively related with closeness to a million-dollar salary in model (i) but unrelated with the million-dollar status in the other two models. However, when the million-dollar variable is interacted with the post-1993 dummy, there exists a significant and negative coefficient in models (i) and (ii). This result suggests that after 1993 increases in salary are smaller, ceteris paribus, for CEOs with salaries of one million dollars or close to one million dollars. When the definition of million-dollar firms is much broader and includes firms with CEOs earning more than one million dollars in salary and bonus at least once during the sample period, the changes in salary also appear to be lower, but the relation is not statistically significant [model (iii)].

4.3. The structure of compensation contracts and the substitution effect

The evidence thus far indicates that some million-dollar firms have responded to the regulations by reducing salaries, but without reducing total

Table 5
Changes in salary before and after the regulations

This table contains the results of quantile (medians) regression analyses to examine differences in changes in salary between CEOs of firms subject to Section 162(m) and CEOs of firms less likely to be affected by Section 162(m). Assets, sales, and compensation have been transformed into logarithms**.** CEO tenure is the number of years the CEO has held the CEO title. Earnings per share (EPS) exclude extraordinary items and discontinued operations and are deflated by the prior year's fiscal year-end stock price. Holding period returns are raw holding period returns. In the first model, the million-dollar variable is equal to one if the CEO's salary in the prior year is greater than or equal to one million dollars and is equal to the CEO's salary in the prior year divided by one million for salaries less than one million dollars. In the second model, the million-dollar variable is a dummy equal to one if the CEO earned more than $900,000 in salary in the preceding year. The $900,000 benchmark is selected because firms paying CEOs in that range are or are close to being subject to 162(m) based on salary alone. In the third model, the million-dollar variable is a dummy that is equal to one if the firm's CEO at any one time during the sample period earned more than one million dollars in salary and bonus. We control for CEO tenure, firm size, and contemporaneous and lagged firm performance. Year-dummies are equal to one if the observation is for the said year and equal to zero otherwise. Regression p-values are in parentheses.

	Dependent variable is Δ Ln(*Salary*)		
Explanatory variables	Model (i) *Semi-continuous MM-\$*	Model (ii) > \$900,000 *MM-\$ dummy*	Model (iii) *salary + bonus MM-\$ dummy*
Constant	0.056	0.067	0.076
	(0.00)	(0.00)	(0.00)
Ln (*CEO tenure*)	−0.012	−0.014	−0.015
	(0.00)	(0.00)	(0.00)
Ln(*Assets*)	0.003	0.001	−0.001
	(0.00)	(0.26)	(0.35)
EPS_t/P_{t-1}	0.001	0.002	0.002
	(0.89)	(0.81)	(0.82)
EPS_{t-1}/P_{t-2}	0.019	0.026	0.028
	(0.33)	(0.06)	(0.07)
ΔLn $(Sales)_t$	0.027	0.028	0.030
	(0.00)	(0.00)	(0.00)
ΔLn $(Sales)_{t-1}$	0.039	0.040	0.040
	(0.00)	(0.00)	(0.00)
Holding period returns$_t$	0.000	0.000	0.001
	(0.90)	(0.92)	(0.72)
Holding period returns$_{t-1}$	0.013	0.014	0.013
	(0.00)	(0.00)	(0.00)
Million-dollar variable	−0.014	−0.003	0.006
	(0.07)	(0.70)	(0.13)

Table 5 (*continued*)

	Dependent variable is Δ Ln(*Salary*)		
Explanatory variables	Model (i) *Semi-continuous* *MM-$*	Model (ii) > $900,000 *MM-$ dummy*	Model (iii) *salary + bonus* *MM-$ dummy*
Post-1993 *dummy * Million-dollar variable*	−0.031 (0.00)	−0.025 (0.00)	−0.005 (0.21)
Year-dummies included	Yes	Yes	Yes
Pseudo R^2	0.04	0.04	0.03
Number of observations	5,270	5,270	5,270

compensation as suggested by our previous discussion. This result suggests that there has been a change in the structure of compensation contracts, with a decrease in the importance of salary. Although the objective of Congress in enacting 162(m) was to reduce excessive compensation, shareholder activists were mostly concerned with enhancing the relation between pay and performance. Hence, the increasing importance of performance sensitive compensation components is also consistent with firms responding to shareholder activism pressures. In this case we can formulate

Proposition 3. Performance sensitive components of compensation, such as bonus and stock-based compensation, have become more important after 1993.

Preliminary evidence on Proposition 3 is provided in Table 1 where one can see that salaries increase at a slower pace than other compensation components. We also make a univariate comparison of compensation structure of firms with CEOs earning more or less than $900,000 and of CEOs with high and low managerial ownership. The regulatory effect should be more important for firms with CEOs earning more than $900,000 and firms with low managerial ownership, because these firms are more subject to shareholder pressure to comply with 162(m) and strengthen the pay for performance relation. Prior research by Mehran (1995) supports this notion with the finding of a negative relation between ownership and stock based compensation, as one might expect if boards set compensation contracts to provide additional incentives to managers with low ownership. Lewellen et al. (1987), and Yermack (1995), on the other hand, do not support this agency-based hypothesis.

The trends we observe, but do not tabulate, are consistent with Proposition 3. They suggest that ex post payments based on performance sensitive components of compensation have become more important over time. At least two major caveats of this comparison exist. First, it is possible that a substitution occurred and the nonsalary portion of compensation was increasing because salary increases were constrained by 162(m). This substitution is consistent with economic theory suggesting that a risk averse agent will require a higher level of compensation if the compensation has become riskier (i.e. performance-based rather than a fixed salary). Alternatively, compensation may appear to be more sensitive to firm performance because the bonus payments are larger due to the improved economic performance after 1993.

We address this first limitation by estimating models similar to the models reported in Table 5, but which include the change in the value of stock-based compensation plus bonus for the contemporaneous and previous years. We do not find a statistically significant relation between changes in salary and the changes in the sum of the annual bonus payments and grants of stock-based compensation and variations of this measure suggesting that our model is not able to capture this substitution effect. We address the second limitation in the next section.

4.4. Measuring the pay for performance sensitivity

Although we report an increase in the importance of performance-dependent components of compensation, we have not analyzed whether the annual pay to performance sensitivity has increased. We formulate

Proposition 4. Increased shareholder scrutiny through enhanced disclosure and 162(m) will lead to an increase in the sensitivity of pay to performance after 1993, especially for firms subject to 162(m).

To test this proposition, we investigate the relation between CEO bonus and total CEO compensation and various measures of firm performance, and examine whether this relation is different for firms likely to be affected by the regulations.

4.4.1. The model

Various models have been used to examine the sensitivity of annual compensation flows to firm performance. Murphy (1985) argues that it is important to control for CEO-specific variables that are constant over time but that vary across firms and solves this problem by including CEO dummies. Murphy (1986) reports that the use of first differences instead of compensation levels implicitly controls for these fixed effects. Joskow et al. (1993), Hubbard and Palia (1995), and Joskow et al. (1996), among others, argue that a fixed effects model is better specified for their panel data. This also seems to be the case for our data, hence we report results using a firm fixed effects model. We also estimate all the models using (1) first differences of the logarithms of compensation in the spirit of Murphy (1985) and Sloan (1993) and the models reported in Table 5; and (2) CEO fixed effects instead of firm fixed effects. Our main results are unchanged with these variations. We also include year dummy variables to control for compensation trends.

We estimate the following firm fixed effects regression model:

$$\begin{aligned}\mathrm{Ln}(Comp_{it}) = {} & \alpha_1 \mathrm{Ln}(CEOtenure_{it}) + \alpha_2 \mathrm{Ln}(Assets_{it}) \\ & + \sum_{j=3}^{8} \alpha_j PerformanceMeasures_i \\ & + Post93dummy * \sum_{j=9}^{14} \alpha_j PerformanceMeasures_i \\ & + Post93dummy * Million\text{-}dollar\ variable \\ & * \sum_{j=15}^{20} \alpha_j PerformanceMeasures_i \\ & + \alpha_{21} Million\text{-}dollar\ variable \\ & + \alpha_{22} Post93 * Million\text{-}dollar\ variable \\ & + Yeardummies + Firmdummies + \varepsilon_{it}. \end{aligned} \tag{2}$$

All compensation figures except for the million dollar variables are in 1992-deflated real dollars. The model's explanatory variables are as in Eq. (1) or interactions thereof. We include interaction variables between the three performance measures, the post-1993 dummy variable, and the three million-dollar firm variables to gauge whether the sensitivity of compensation to performance increases after 1993, especially for million-dollar firms. The year t dummy variables are equal to one for compensation in year t and equal to zero all other years. In models (v) and (vi), we do not include the million-dollar variable by itself or its interaction with the post-1993 dummy because under this definition the firm either has a dummy value of one or zero for the entire sample period and can therefore not be included in a firm fixed effects model.

In our models, bonus is a linear function of performance. Holthausen et al. (1995) use nonpublic survey data from a consulting firm and document that many bonus plans are fixed-target plans in which executives do not receive any payoff until they reach a lower bound of the performance measure. Between the lower and the higher bound, the bonus increases linearly with the performance measure. Beyond the higher bound and the maximum bonus, additional performance is not reflected in the bonus. These features can reduce the explanatory power of our model.

4.4.2. The results

Table 6 presents our results. The dependent variable for models (i), (iii), and (v) is the logarithm of bonus and for models (ii), (iv), and (vi) the dependent variable is the logarithm of total compensation.[7,8] In all models, CEO tenure is unrelated to bonus payments and total compensation, however, firm size, measured by the natural logarithm of total assets, is positively related to total compensation but unrelated to bonus payouts.

Focusing on the first six performance coefficients, the bonus payments are positively related to all three contemporaneous performance measures and also to lagged returns. Total compensation is only positively related to contemporaneous profitability, as measured by deflated earnings per share. Next, we examine the relation with performance measures post-1993. The first four coefficients examine the sensitivity of bonus and total pay to contemporaneous and lagged profitability. The next four coefficients examine the sensitivity of bonus and total pay to contemporaneous and lagged sales growth. There is a decline in the relation between bonus payouts and total compensation and profitability after 1993, but this decline in sensitivity for bonus payouts does not exist or is less pronounced for million-dollar firms. The sensitivity with lagged earnings is less clear. In model (i) there is an increasing positive relation with lagged earnings, but less so as firms approach the million-dollar status. Bonus payouts are less sensitive to contemporaneous sales growth after 1993 across all the million-dollar definitions, and there is also an increase in the sensitivity to lagged sales for total compensation in model (vi).

The most important performance measure to shareholders is stock returns and the changes in sensitivity post-1993 are much clearer with respect to the sensitivity to stock returns. Specifically, there is an increase in the sensitivity to contemporaneous stock returns post-1993 for both bonus payouts and total compensation (significant in five of six models) and this increase is even more pronounced for the million-dollar firms (significant in four of six models). For lagged stock returns, there is also an increase in the sensitivity after 1993, especially for million-dollar firms (significant in four of the six models). This relation does not exist in models (iii) and (iv) where the million-dollar variable

[7] Using the natural logarithm of bonus leads to computational complications because a small fraction of bonuses are equal to zero (approximately 19%). The fraction of bonuses equal to zero is similar for each year in the sample period. When the bonus grant is equal to zero, we assign a discretionary value of one to those payments, so that the logarithm is zero. If we assign other values, such as 0.001 or 1,000 instead of one, the magnitude of the coefficients changes significantly and the signs of the coefficients on tenure and ln (sales) are reversed between the choice of 0.001 and one and the choice of 1,000. Additionally, the explanatory power of the model is larger when we use 1,000 instead of 0.001 and one for bonuses with values equal to zero.

[8] There are fewer observations in models (i)–(iv) because the million-dollar variable definition requires knowledge of the cash compensation in the preceding year. As a result we lose some 1991 or 1992 observations.

Table 6
Sensitivity of compensation to firm performance following Section 162(m)

This table contains the results of a firm fixed effects regression on the CEO bonus and on total CEO compensation as dependent variables. We use time series data for firms from the *ExecuComp* database over 1991–1997. Assets, sales, and compensation have been transformed into logarithms. CEO tenure is the number of years the CEO has held the CEO title. Earnings per share (EPS) exclude extraordinary items and discontinued operations and are deflated by the prior year's fiscal year-end stock price. Holding period returns are raw holding period returns. In the first two models, the million-dollar variable is equal to one if the CEO's salary in the prior year is greater than or equal to one million dollars and is equal to the CEO's salary in the prior year divided by one million for salaries less than one million dollars. In the next two models, the million-dollar variable is a dummy equal to one if the CEO earned more than $900,000 in salary in the preceding year. The $900,000 benchmark is selected because firms paying CEOs in that range are or are close to being subject to 162(m) based on salary alone. In the last two models, the million-dollar variable is a dummy that is equal to one if the firm's CEO at any one time during the sample period earned more than one million dollars in salary and bonus. For each pair of models, the first model analyzes the CEO bonus and the second model focuses on total CEO compensation. Compensation data are in CPI-deflated 1992 constant dollars. Year-dummies are equal to one if the observation is for the said year and equal to zero otherwise. Regression *p*-values are in parentheses. For models (ii), (iv), and (vi) the sample consists of a smaller set of firms for which option values are available.

	Dependent variable					
	Ln(*Bonus*)	Ln(*Total pay*)	Ln(*Bonus*)	Ln(*Total pay*)	Ln(*Bonus*)	Ln(*Total pay*)
Explanatory variables	Model (i) *Semi-continuous*	Model (ii) *MM-$*	Model (iii) > $900,000	Model (iv) *MM-$ dummy*	Model (v) *Salary + bonus*	Model (vi) *MM-$ dummy*
Constant	3.160	4.742	2.952	4.688	3.483	4.481
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Ln (*CEO tenure*)	−0.038	0.023	−0.066	0.036	−0.055	0.010
	(0.70)	(0.68)	(0.49)	(0.48)	(0.33)	(0.64)
Ln (*Assets*)	0.182	0.306	0.138	0.315	0.055	0.349
	(0.17)	(0.00)	(0.29)	(0.00)	(0.65)	(0.00)
EPS_t/P_{t-1}	6.648	1.158	6.740	1.138	6.201	0.885
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
EPS_{t-1}/P_{t-2}	−0.227	0.085	−0.263	0.091	0.015	0.007
	(0.46)	(0.25)	(0.41)	(0.22)	(0.96)	(0.91)

Table 6 (*continued*)

Explanatory variables	Ln(*Bonus*)	Ln(*Total pay*)	Ln(*Bonus*)	Ln(*Total pay*)	Ln(*Bonus*)	Ln(*Total pay*)
	Dependent variable					
	Model (i)	Model (ii)	Model (iii)	Model (iv)	Model (v)	Model (vi)
	Semi-continuous MM-$		> $900,000 *MM-$ dummy*		*Salary + bonus MM-$ dummy*	
ΔLn (*Sales*)$_t$	1.340	0.089	1.384	0.108	0.959	0.107
	(0.00)	(0.50)	(0.00)	(0.41)	(0.00)	(0.24)
ΔLn (*Sales*)$_{t-1}$	0.031	−0.075	0.053	−0.052	0.004	−0.057
	(0.88)	(0.32)	(0.78)	(0.50)	(0.98)	(0.39)
Holding period returns$_t$	0.217	0.025	0.227	−0.013	0.310	−0.000
	(0.03)	(0.54)	(0.03)	(0.77)	(0.00)	(0.99)
Holding period returns$_{t-1}$	0.175	0.006	0.178	0.008	0.206	0.016
	(0.02)	(0.82)	(0.02)	(0.76)	(0.00)	(0.29)
Post-1993 *dummy times* EPS_t/P_{t-1}	−5.135	−0.978	−3.842	−1.002	−4.196	−0.637
	(0.00)	(0.02)	(0.00)	(0.00)	(0.00)	(0.04)
Post-1993 *dummy times* EPS_t/P_{t-1} *times million dollar variable*	2.923	−0.185	6.724	−1.103	4.339	0.393
	(0.22)	(0.74)	(0.02)	(0.20)	(0.00)	(0.20)
Post-1993 *dummy times* EPS_{t-1}/P_{t-2}	3.866	0.253	0.247	0.167	0.682	−0.153
	(0.00)	(0.62)	(0.72)	(0.39)	(0.26)	(0.34)
Post-1993 *dummy times* EPS_{t-1}/P_{t-2} *times million dollar variable*	−8.168	−0.361	−3.796	−0.566	−2.148	0.452
	(0.00)	(0.70)	(0.11)	(0.52)	(0.01)	(0.11)
Post-1993 *dummy times* ΔLn (*Sales*$_t$)	−0.670	0.063	−1.021	−0.071	−0.624	0.001
	(0.08)	(0.72)	(0.00)	(0.61)	(0.02)	(0.99)
Post-1993 *dummy times* ΔLn (*Sales*$_t$) *times million dollar variable*	−0.732	−0.254	0.222	−0.092	−0.085	−0.150
	(0.27)	(0.37)	(0.75)	(0.73)	(0.76)	(0.14)

Post-1993 *dummy times* ΔLn ($Sales_{t-1}$)	−0.111	0.070	−0.062	0.048	0.012	0.157
	(0.76)	(0.63)	(0.80)	(0.60)	(0.96)	(0.06)
Post-1993 *dummy times* ΔLn ($Sales_{t-1}$) *times million dollar variable*	0.203	0.053	0.314	0.054	0.133	−0.223
	(0.78)	(0.84)	(0.72)	(0.81)	(0.67)	(0.03)
Post-1993 *dummy times holding period returns*$_t$	0.398	0.105	1.008	0.248	0.758	0.116
	(0.06)	(0.20)	(0.00)	(0.00)	(0.00)	(0.02)
Post-1993 *dummy times returns*$_t$ *times million dollar variable*	1.453	0.383	0.226	0.248	0.253	0.219
	(0.00)	(0.01)	(0.59)	(0.07)	(0.11)	(0.00)
Post-1993 *dummy times holding period returns*$_{t-1}$	−0.095	0.015	0.279	0.150	0.161	0.067
	(0.53)	(0.81)	(0.00)	(0.00)	(0.08)	(0.03)
Post-1993 *dummy times returns*$_{t-1}$ *times million dollar variable*	0.908	0.368	−0.080	0.140	0.336	0.133
	(0.00)	(0.01)	(0.84)	(0.32)	(0.01)	(0.02)
Million-dollar variable	−1.322	0.110	−0.640	−0.053	—	—
	(0.10)	(0.70)	(0.04)	(0.55)		
Post-1993 *dummy times million-dollar variable*	−0.032	0.194	−0.044	0.173	—	—
	(0.93)	(0.14)	(0.90)	(0.10)		
1993-*dummy*	0.386	0.021	0.360	0.033	0.383	0.052
	(0.00)	(0.57)	(0.00)	(0.35)	(0.00)	(0.07)
1994-*dummy*	0.868	−0.022	0.806	0.091	0.698	0.081
	(0.00)	(0.80)	(0.00)	(0.03)	(0.00)	(0.02)
1995-*dummy*	0.550	−0.013	0.469	0.110	0.375	0.096
	(0.01)	(0.88)	(0.00)	(0.01)	(0.00)	(0.01)
1996-*dummy*	0.667	0.065	0.598	0.206	0.458	0.214
	(0.00)	(0.47)	(0.00)	(0.00)	(0.00)	(0.00)
1997-*dummy*	0.750	0.210	0.698	0.358	0.590	0.344
	(0.00)	(0.03)	(0.00)	(0.00)	(0.00)	(0.00)
Adjusted R^2	0.61	0.76	0.61	0.76	0.60	0.76
Regression p-value	0.00	0.00	0.00	0.00	0.00	0.00
Sample size	5,270	5,062	5,270	5,062	6,166	5,853

is only one for CEOs earning more than $900,000 of salary. It is possible that this definition is not broad enough to identify all the firms that are concerned about 162(m), or that the lack of significant relation is due to the smaller sample of firms defined as million-dollar firms with this definition.

The negative coefficient on the million-dollar variable in the bonus models suggests that bonus payouts have been lower for million-dollar firms when other factors including performance are taken into account. The positive coefficients on the million-dollar and post-1993 indicator variable suggest marginally higher total compensation payouts for million-dollar firms after 1993. Together with the lower salary increases for million-dollar firms post-1993, this result suggests that there has been some substitution from salary to other compensation components (i.e. stock options or restricted stock) for million-dollar firms. These coefficients are not, however, significant at conventional levels of statistical significance.

One important conclusion of our analysis of the sensitivity of CEO bonus and total compensation to performance is that there is a stronger relation between bonus payments and/or total pay and stock returns post-1993, and that this relation is significantly more pronounced for firms that are more likely to be affected by the regulations. In addition, this relation holds for both contemporaneous and lagged returns. The existing relation with lagged returns is understandable given the timing when some firms determine the bonus payouts and total pay.

Murphy (1999) and Hall (1999) document that there is a mechanical relation between option grants (included in the total compensation measure) and rising stock markets because 40% of the companies grant options on a fixed share basis. Granting a fixed number of shares annually will mean that rising stock values will mechanically lead to stock option grants with higher values. We do not believe that this mechanical relation explains our findings because we find a strong increased sensitivity between bonus cash payouts (which do not include option payouts) and stock returns in models (i), (iii), and (v).

5. Measures of changes in CEO wealth for changes in shareholder wealth

Thus far, we have focused on the sensitivity of annual compensation flows (i.e., salary, bonus, restricted stock, stock options, long-term incentive plan payouts, and other compensation) to firm performance. We have taken this approach because compensation critics have concentrated on these annual flows and because this is the flow that was apparently targeted by the regulations. Jensen and Murphy (1990), Hall and Liebman (1998), Murphy (1999), among others, show that CEO equity and option ownership in the firm accounts for the bulk of the sensitivity of CEO wealth to firm performance.

While our prior results indicate that compensation structure and the compensation to performance sensitivity of million-dollar firms have been affected by the regulations, a crucial question to determine the economic impact of the regulations is whether these effects translate into a higher sensitivity of CEO wealth to shareholder wealth. To examine this issue, we formulate

Proposition 5. After the regulations, the change in CEO wealth per dollar change in shareholder wealth increases, especially for million-dollar firms.

To study the evolution of CEO ownership incentives, we first examine the bottom two rows of Table 1, in which we report the percentage of CEO stock and option ownership for our sample firms. Both means and medians suggest that CEO stock ownership has been relatively stable or even declined during the sample period. In contrast, the mean (median) stock option ownership increases from 1.0% (0.4%) in 1992 to 1.1% (0.7%) in 1997. This result is consistent with Ofek and Yermack (2000) who find that executives often divest shares following the exercise of stock options.

To further investigate Proposition 5, we use an approach similar to Hall and Liebman (1998) and present three measures of the relation between CEO wealth changes and shareholder wealth changes. We estimate CEO wealth using a methodology that combines the methodology of Hall and Liebman (1998) and Murphy (1999). For each CEO, we calculate an implied "compensation" or CEO wealth change for a given level of performance in the following manner. First, we estimate the implied salary and bonus for each firm by estimating an implicit cash compensation-for-performance elasticity for each year and for each subgroup using the quantiles regression model $\Delta\ln(salary + bonus) = \ln(1 + return)$. The year and subgroup elasticity (defined by million-dollar variable) is then converted into a firm-specific implied salary and bonus by multiplying by the return at a given performance level and the CEO's salary and bonus for the prior year (Murphy, 1999). The implicit elasticities are similar to those calculated by Murphy and Hall and Liebman and range from 0.26 to 0.52 across subgroups. Second, we compute the change in the value of the CEO's option portfolio using a delta of 0.6. Third, we estimate the change in the value of the common stock and restricted stock portfolio held by each CEO. Finally, we add the value of the median level of restricted stock grants, option grants and other annual compensation for the entire sample of firms, which translates into holding constant each of these components across all firms. One major difference in our calculation of each CEO's compensation compared to Hall and Liebman is that the salary and bonus payments are not held constant across firms in our calculations of CEO wealth.

To ensure that our results are not driven by changes in the sample composition, we only use the firms that have complete information for both 1993 and 1996 in calculating the three sensitivity measures. The first measure, the percent change in compensation, is defined as the change in the CEO's wealth for a typical change in firm performance, defined as an increase from the 50th percentile of stock returns to the 70th percentile.[9] Our second measure is an elasticity, defined as the percent change in compensation calculated above divided by the percent change in market value or difference in total shareholder return from the 50th to the 70th percentile in that year. Our third and final measure of the relation between CEO wealth and performance is the dollar change in CEO wealth for a $1,000 change in market value, called the *Jensen-Murphy statistic*. We calculate the Jensen-Murphy statistic as the sum of (1) the common and restricted stock sensitivity, which is the fractional ownership of common and restricted stock, (2) the options sensitivity, which is the number of options held multiplied by an implied delta of 0.6 and divided by the number of shares outstanding, and (3) the implicit cash compensation sensitivity, calculated as the cash compensation elasticity described above, multiplied by the CEO's salary and bonus and divided by the market value of the firm. We compare these three measures for million-dollar firms and non-million-dollar firms for wealth changes in 1993 (pre-regulation) and wealth changes in 1996 (post-regulation) and report these results in Table 7.

The median percent change in wealth for a CEO and the change in elasticity of compensation as performance moves from the 50th to the 70th percentile are similar to the results reported by Hall and Liebman (1998). While there is little difference across the subgroups, the percent change in compensation decreases from 1993 to 1996 whereas the elasticity increases from 1993 to 1996. One explanation of this result is the difference in shareholder returns from the 50th to the 70th percentile for the two years (18 percentage points in 1993 and 16.3 percentage points in 1996). The inverse relation between size and the Jensen-Murphy statistic previously documented by Murphy (1999), among others, is also evident in the results in Table 7. The Jensen-Murphy statistic is significantly lower for million-dollar firms, which is expected because CEOs with higher salaries tend to be from larger firms and large-firm CEOs tend to have lower fractional ownership stakes. Overall, we do not find any clear trend in the univariate analysis of these sensitivity measures.

[9] This definition of typical firm performance follows Hall and Liebman (1998), and using the distribution of stock returns for all of the *ExecuComp* firms in our sample, the stock return at the 50th percentile in 1993 (1996) was 15.8% (16.3%) and at the 70th percentile was 33.8% (32.6%). We use 1996 as our post-regulation comparison year because of the similar distribution of returns.

To further examine the change in these wealth to performance measures, we estimate the following regression model

$$\text{Change in the wealth to performance sensitivity measure from 1993 to 1996 for each firm} = \alpha_0 + \alpha_1 \ln(CEO\,tenure)_{93} + \alpha_2 \ln(assets)_{93} + \alpha_3\, CEO\ turnover\ dummy + \alpha_4 \ln(CEO\,dollar\,ownership)_{93} + \alpha_5\, \$MMFirm + \varepsilon. \quad (3)$$

Our results are reported in Table 8. In models (i)–(iii), the dependent variable is the change in the percent change from 1993 to 1996. In models (iv)–(vi), the dependent variable is the change in the elasticity measure. Finally, in models (vii)–(ix), the dependent variable is the change in the Jensen-Murphy statistic. For each series of three models, we use the three million-dollar proxy variables we have used in our previous regressions.

In all but two of the models, the constant is positive and significant suggesting that there has been an overall increase in the CEO wealth to performance sensitivity. We also find that CEO tenure does not affect the change in CEO wealth to performance sensitivity from 1993 to 1996. Size, however, seems to have a positive impact on the increase in the sensitivity of wealth to performance from 1993 to 1996. This is consistent with the notion that large firms are more likely to be pressured to increase CEO incentives. As expected, CEO turnover leads to a decline in the CEO's wealth to performance sensitivity over the corresponding period. This result suggests that new CEOs do not immediately have wealth to performance sensitivities that are close to the outgoing CEO's sensitivity, which is consistent with new CEOs having lower levels of ownership of both stock and options than exiting CEOs. This relation is not significant, however, in the last three regressions in which the dependent variable is the Jensen-Murphy statistic. We also control for CEO ownership, measured as the logarithm of the dollar value of the CEO's ownership in 1993, and find that higher levels of ownership result in smaller changes in sensitivity from 1993 to 1996. This measure of CEO ownership is suggestive of how much the CEO has at stake (relative to her/his unknown total portfolio), and the results are consistent with the argument that firms with high CEO ownership have less pressure to increase CEO incentives by further aligning CEO and shareholder wealth. We also estimate the models with CEO ownership as a percentage of total firm ownership. The results are qualitatively unchanged with this measure of CEO ownership.

The most important results for our analysis are the coefficients and significance levels of the million-dollar variables. For all but one (model ix) of the nine models, we find a significant positive relation between the million-dollar variable and the change in sensitivity. This result indicates that controlling for other potentially important factors, the sensitivity of CEO

Table 7
Measures of the sensitivity of CEO wealth to changes in shareholder wealth

In this table, we compare medians of the three measures of the sensitivity of changes in CEO wealth to changes in shareholder wealth. The three measures are: (i) the percent change in compensation, defined as the change in the CEO's wealth for a typical change in firm performance, which we define as an increase from the 50th percentile of stock returns to the 70th percentile (following Hall and Liebman, 1998), (ii) the elasticity, defined as the percent change in compensation calculated above divided by the percent change in market value or difference in total shareholder return from the 50th to the 70th percentile in that year, and (iii) the Jensen-Murphy statistic, defined as the dollar change in compensation for a $1,000 change in market value. We compare these three measures for firms that are classified as million-dollar firms using two of the million-dollar variables as defined previously. We do not use the semicontinuous million-dollar variable because it is not an indicator variable.

	Percent change in compensation from the 50th to 70th percentile		Elasticity		Jensen-Murphy statistic	
	1993	1996	1993	1996	1993	1996
Salary in 1993 < \$900,000 ($n = 820$)	0.78	0.71	4.33	4.35	16.02	16.42
Salary in 1993 > \$900,000 ($n = 71$)	0.88	0.83	4.85	5.09	7.08	7.43
Salary and bonus < \$1,000,000 ($n = 404$)	0.80	0.67	4.46	4.09	24.65	21.43
Salary and bonus > \$1,000,000 ($n = 487$)	0.78	0.75	4.33	4.60	11.36	11.38

wealth to firm performance has increased significantly for firms that are more likely to be affected by 162(m) relative to other firms. Overall, these results are consistent with our prior results on annual compensation flows.

6. Additional tests and robustness checks

6.1. Ownership or shareholder targeting

We perform various tests to examine whether our results are sensitive to measurement method or sample selection. Thus far we have focused on the subset of million-dollar firms because they are more likely to be affected by the regulations. We now examine whether the million-dollar effect is due to the fact that million-dollar firms are large firms with common governance characteristics that make them more sensitive to shareholder pressures relating to compensation. Core et al. (1999), among others, document that governance structure affects CEO compensation. We explore the characteristics of million-dollar firms using an untabulated logistic model and confirm that million-dollar firms tend to be large firms with low CEO ownership. Firms with low CEO ownership are firms where the CEO has less control or where the board is more likely to face investor pressures to increase the alignment between CEO and shareholder incentives.

Next, we estimate the models from Table 6 and add interaction variables for (i) firms with low CEO ownership (a dummy equal to one if the CEO's ownership is lower than the median CEO ownership for 1993 and equal to zero otherwise), and (ii) firms that have been targeted by shareholder activists using various targeting definitions with data from Strickland et al. (1996) and Wahal (1996). The sensitivity to stock returns increases both for low ownership firms and for both definitions of million-dollar firms after 1993 compared to firms with high ownership or firms that are not million-dollar firms with high ownership. A similar yet weaker pattern is also observable for the sensitivity to EPS. Overall, these results suggest that sensitivities to performance measures increased after 1993 both for million-dollar and low ownership firms. This result is consistent with the notion that 162(m) affected the pay for performance sensitivity and that boards of firms with low CEO ownership increased the pay for performance sensitivity of compensation contracts to align CEO and shareholder interests. We do not find such a result for interactions with the shareholder activist targeting dummy. Overall, our results show that although the governance variables matter, they do not affect our prior results, namely that million-dollar firms have increased the sensitivity of compensation to performance beyond what other firms have done following the enactment of 162(m). These results are not tabulated for brevity.

Table 8
Changes in the wealth to performance sensitivity of CEOs following Section 162(m)

This table contains the results of regression with robust standard errors on three measures of the sensitivity of the change in the CEO's wealth to changes in firm performance. The three empirical proxies are defined in the heading to Table 7. The change is the increase for each firm in the measure of CEO wealth to firm performance sensitivity from 1993 to 1996. The control variables are the logarithm of CEO tenure, where CEO tenure is the number of years the CEO has held the CEO title; the logarithm of the firm's assets; a dummy equal to one if there was a change in the CEO during that period; and the logarithm of the dollar value of the CEO's ownership. The key test variable is designed to identify firms that are more likely to be subject to 162(m). We use the same three million-dollar variables that are used in Tables 5 and 6. These variables are explained in detail in the heading to Table 5. Regression p-values are in parentheses and are calculated using White-corrected standard errors.

	Dependent variable is								
	Change in the percent change			Change in the elasticity			Change in the Jensen-Murphy statistic		
Explanatory variables	Model (i)	Model (ii)	Model (iii)	Model (iv)	Model (v)	Model (vi)	Model (vii)	Model (viii)	Model (ix)
Constant	0.068	0.062	0.097	0.186	0.158	0.375	0.036	0.026	0.022
	(0.07)	(0.09)	(0.01)	(0.38)	(0.46)	(0.07)	(0.02)	(0.04)	(0.06)
Ln (*CEO tenure*)	−0.002	−0.00002	−0.002	−0.012	−0.001	−0.011	−0.0005	0.001	0.001
	(0.80)	(0.99)	(0.84)	(0.80)	(0.98)	(0.81)	(0.83)	(0.64)	(0.61)
Ln (*Assets*)	0.024	0.029	0.021	0.135	0.157	0.112	−0.001	0.002	0.002
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.48)	(0.00)	(0.00)
CEO turnover dummy	−0.063	−0.061	−0.058	−0.393	−0.380	−0.362	−0.002	−0.001	−0.001
	(0.04)	(0.04)	(0.05)	(0.03)	(0.03)	(0.04)	(0.79)	(0.89)	(0.88)
Ln (*CEO dollar ownership*)	−0.036	−0.036	−0.038	−0.139	−0.139	−0.147	−0.006	−0.005	−0.005
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Semicontinuous Million-dollar variable	0.065	—	—	0.341	—	—	0.042	—	—
	(0.03)			(0.05)			(0.00)		
> \$900,000 *Million-dollar variable*	—	0.032	—	—	0.171	—	—	0.010	—
		(0.07)			(0.09)			(0.01)	
Salary + bonus Million-dollar variable	—	—	0.060	—	—	0.353	—	—	0.002
			(0.00)			(0.00)			(0.49)
R^2	0.18	0.26	0.29	0.18	0.18	0.21	0.12	0.08	0.08
Regression p-value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sample size	684	684	684	684	684	684	684	684	684

6.2. Firms that do not pay taxes?

If 162(m) has an effect on the pay for performance sensitivity, then non-tax-paying firms should be less sensitive to its influence. To examine this assumption, we identify 310 firm-years, representing 169 different firms, with zero tax rates according to Graham's (1996) trichotomous tax rate approximations. We repeat the tests presented in Table 6 using dummies and interactions for the tax-paying status. We do not find an effect of the tax-paying status for bonus payments sensitivity, but for total compensation we find a stronger sensitivity to stock returns after 1993 for million-dollar tax-paying firms. This result is consistent with the notion that tax-paying firms should be more sensitive to 162(m) than nontax-paying firms.

6.3. Variations in the measure of the firms likely to be affected by 162(m)

We present all our results with three very different proxies designed to capture how sensitive firms would be to the impact of 162(m). The selection of the correct variable(s) to identify million-dollar firms is essential to perform robustness tests of the impact of 162(m) on compensation levels and on the relation between pay and performance. Rose and Wolfram (2000) discuss how the selection of some of these variables could potentially bias results of researchers in this area. For example, our third measure of million-dollar firms is an indicator variable equal to one for firms with CEOs earning annual cash compensation (including salary and bonus) of more than one million dollars at least once over the 1992–1997 period. Our intention is to broadly include all firms that during the sample period have or could have been affected by 162(m). Discussing a prior version of this paper, Rose and Wolfram point out that this identification includes firms that earned a million dollars only at the end of the period of interest.

To verify the robustness of our results, we have estimated our results with a wide variety of million-dollar variables and present three of the most representative variables. Our results are similar in nature for all three definitions (although the significance of the coefficients differs). Specifically, Table 5 documents lower increases in salary, ceteris paribus, for million-dollar firms, and Table 6 documents that an increase in the sensitivity of bonus payouts and total compensation to stock returns for million-dollar firms after 1993. Finally, Table 8 shows that for million-dollar firms there is an increase in the sensitivity of CEO wealth to performance post-1993 after controlling for other factors that affect CEO incentives.

We also use several other million-dollar definitions. For instance, for one definition we only include firms with CEOs who have a salary that is higher than one million dollars. This change in the definition does not change the qualitative nature of our results. We also broaden the million-dollar dummy definition to include firms in which the CEO has earned at least \$800,000 or \$750,000 in salary and find that our results on the million-dollar dummy are weakened and eventually disappear, as expected. The further away a firm is from a million-dollar salary, the less sensitive it will be to 162(m).

6.4. Poor performing firms

Corporate performance has been strong over the sample period. Is the positive relation between stock returns and compensation due to this positive performance? To examine this issue, we estimate the models in Table 6 for poor performers only. We define poor performers as firms with at least two years of negative stock returns over the sample period, or more restrictively with two years of negative stock returns after 1993. Although the coefficients and significance of our results differ from Table 6, we still find an increase in the sensitivity of bonus to stock returns for all firms, and especially for million-dollar firms after 1993, suggesting that the increase in pay for performance sensitivity documented in Table 6 also applies to poorly performing firms.

7. Concluding comments

The objectives behind the adoption of the new SEC Compensation Disclosure Rules and Section 162(m) were to reduce excessive CEO compensation levels and strengthen the pay for performance relation. We formulate five propositions to test the effect of these compensation regulations using 1991–1997 CEO compensation data. We find that while the regulations have not achieved the objective of reducing CEO compensation growth, they appear to have had a statistically and economically significant impact on the compensation structure of firms most likely to be affected by the regulations.

We also document an impact on the level and structure of compensation for an easily identifiable subset of firms that reduce CEO salaries from more than one million dollars to one million dollars or below. Twenty-three of the 25 firms doing so refer to 162(m), however, total compensation still increases for a majority of these firms. We also find an increasing relation between stock returns and both bonus payments and total compensation after 1993 for all firms. Consistent with the law's effect, this finding is especially pronounced for million-dollar firms. Robustness tests show that the increased pay for performance relation is not driven by surging stock option grants, the definition of million-dollar firms, or CEO ownership.

To further understand the effect of 162(m) and the disclosure regulation on our large sample of firms, we also examine the sensitivity of three measures of CEO's wealth to firm value. Again, we use our three alternative measures of a firm's million-dollar status and find that, relative to other firms, there has been an increase in the sensitivity of CEO wealth to shareholder wealth from 1993 to 1996 for million-dollar firms after controlling for other factors that affect CEO incentives. In sum, our results suggest that compensation committees have taken the regulatory environment into account, and that these regulations have had a real economic impact on compensation and on the overall pay for performance relation. The reaction of firms with a pay for performance emphasis in their compensation policy to the currently changing economic conditions will generate interesting observations on the durability of this phenomenon.

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